

SANTHOSH ANDAVAR

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PROFESSIONAL SUMMARY

Full-stack software engineer and recent Computer Science graduate (BS, GPA 3.6) with strong internship experience designing, building, and operating production web and mobile applications across two software engineering internships. Comfortable across the stack — Java, Python, TypeScript, React, Node.js, Spring Boot, and FastAPI — with REST APIs backed by relational and NoSQL databases. Hands-on with AWS, Docker, Kubernetes, CI/CD pipelines, microservices, and modern AI / Generative AI workflows (LLMs, RAG, embeddings). Collaborative, fast learner, and effective in Agile Scrum teams from design through deployment and on-call support.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML5, CSS3, Bash, C++

Backend: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Node.js, Express.js, NestJS, Python, FastAPI, Flask, Django, REST APIs, GraphQL, Microservices, Event-Driven Architecture

Frontend: React, Next.js, React Hooks, React Router, Redux, Angular, Tailwind CSS, Responsive Web Design, Accessibility

Mobile: React Native, Expo, Cross-Platform App Development

Databases: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, MongoDB, DynamoDB, Redis, pgvector, Schema Design, Query Optimization, Indexing, Stored Procedures

Cloud: AWS, EC2, ECS, S3, Lambda, RDS, DynamoDB, API Gateway, SQS, SNS, CloudWatch, IAM, KMS, Bedrock, Azure, App Services, Functions, GCP

DevOps & CI/CD: Git, GitHub, GitLab, GitHub Actions, Jenkins, Docker, Kubernetes, Helm, Terraform, CI/CD Pipelines, Feature Flags

AI / GenAI: LLM APIs, OpenAI, Anthropic Claude, Retrieval-Augmented Generation, Embeddings, Vector Databases, pgvector, Prompt Engineering, LangChain, Whisper, AI-Assisted Dev Tools, GitHub Copilot, Claude, Cursor

Testing & Quality: JUnit 5, Mockito, pytest, Jest, React Testing Library, Cypress, Postman, Unit Testing, Integration Testing, Contract Testing

Security: OAuth 2.0, OpenID Connect, JWT, Authentication, Authorization, Secure Coding, Input Validation, OWASP Top 10

Observability: CloudWatch, Application Insights, Log/Metric Analysis, Alerting, On-Call Rotation, Incident Response

Methodologies: Agile, Scrum, Sprint Planning, Code Reviews, Pair Programming, SDLC, Technical Documentation

PROFESSIONAL EXPERIENCE

CrowdStrike

Apr 2025 – May 2026

Software Engineer Intern

USA

- Contributed to full-stack features in Java (Spring Boot), TypeScript/Node.js, and Python (FastAPI) for a high-scale production platform under senior-engineer mentorship.
- Built and consumed REST APIs across a microservice architecture, with persistence in PostgreSQL, DynamoDB, and Redis, deployed to AWS via Docker and Kubernetes (EKS).
- Developed React + TypeScript frontends with reusable component libraries, server-side rendering with Next.js, and accessibility-first patterns.
- Contributed to AI / GenAI features including a RAG assistant over internal documentation using OpenAI / Anthropic Claude APIs, embeddings, and pgvector.
- Wrote unit and integration tests with JUnit 5, Mockito, Jest, and Cypress, helping raise coverage to 85% and reducing production regressions.
- Worked with CI/CD pipelines via GitHub Actions and Jenkins, including feature flags, safe-deploy patterns, and rollback playbooks.
- Shadowed on-call rotations, performed defect isolation with CloudWatch logs/metrics, and supported incident follow-ups and post-mortems.

- Participated in design discussions and code reviews, and contributed to internal knowledge-sharing on full-stack patterns and GenAI integration.

Cleveland Clinic

Software Engineer Intern

Apr 2023 – Mar 2024

Cleveland, OH

- Built full-stack healthcare applications using React, TypeScript, Node.js, and Python, supporting clinical and administrative workflows used by thousands of internal users.
- Developed REST APIs in Spring Boot and FastAPI, backed by PostgreSQL and Microsoft SQL Server, with secure auth (OAuth 2.0, JWT) and HIPAA-aware data handling.
- Used Git, GitHub, Jenkins, Docker, and AWS (EC2, S3, RDS, CloudWatch) for source control, CI/CD, and deployment under senior-engineer mentorship.
- Participated in Agile Scrum ceremonies including sprint planning, stand-ups, retrospectives, and code reviews, strengthening fundamentals and shipping discipline.
- Wrote unit tests with Jest, JUnit, and pytest, and contributed documentation that improved onboarding for new interns and engineers.
- Debugged cross-system integrations and learned production monitoring and alerting patterns under senior-engineer guidance.

PROJECTS

MyVisaPrep AI

- Built a GenAI-powered mock interview platform that helps international applicants prepare for U.S. visa interviews with realistic, on-demand sessions.
- Implemented retrieval-augmented generation (RAG) over a curated visa Q&A corpus with embeddings and pgvector, generating context-aware questions and dynamic follow-ups based on user answers.
- Developed a real-time voice analysis pipeline using Whisper plus audio features (pitch, pacing, hesitation, filler words) to compute a confidence score and personalized feedback.

FYX (AI Fitness & Nutrition App)

- Built a startup-grade AI fitness & nutrition mobile app with body-composition analysis, personalized workout/meal planning, and a \$50 Stripe-based accountability loop with refund logic.
- Architected a separate Python AI microservice (FastAPI + PyTorch + OpenCV) for body classification, workout-form pose analysis, and food recognition, called by a NestJS backend over REST.
- Designed a privacy-first encrypted media pipeline: AWS S3 SSE-KMS, pre-signed upload URLs, role-based access, and zero human access to raw body images.
- Implemented an automated accountability engine with BullMQ on Redis and Stripe charge/refund webhooks, plus a Next.js admin dashboard and GitHub Actions CI/CD across dev/staging/prod.

EDUCATION

Cleveland State University

Bachelor of Science, Computer Science

Aug 2021 – May 2025

Cleveland, OH

- GPA: 3.6 / 4.0
- Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Databases, Distributed Systems, Cloud Computing, Operating Systems, Computer Networks, Machine Learning

CERTIFICATIONS

- AWS Certified Solutions Architect
- CompTIA Security+
- CompTIA Network+
- Google Cybersecurity Professional Certificate – Coursera (Google)
- Python Programming – Coursera (University of Pennsylvania)
- Cloud Computing – Coursera (IBM)
- Full-Stack Development – Coursera (IBM)
- Software Development – Coursera (Amazon)